

Goal: See **why** we might model a situation with a graph, and learn some vocabulary.

1. **vertex** (plural: **vertices**):

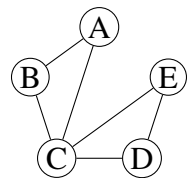
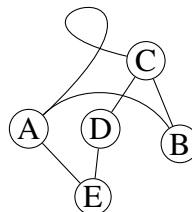
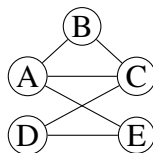
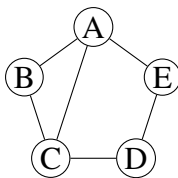
edge:

situation	vertices are...	edges are...
The road system connecting some towns in Alaska.		
The streets in a neighborhood that a snowplow has to clear after a storm.		
A group of people, and who is friends with whom.		
Classes offered at this university, and the students that take them.		

2. We have information about five people and who is friends with whom. Draw a graph.

person	friends with
Rosa	Sam, Tavi
Sam	Rosa, Tavi, Uma
Tavi	Rosa, Sam
Uma	Sam, Wes
Wes	Uma

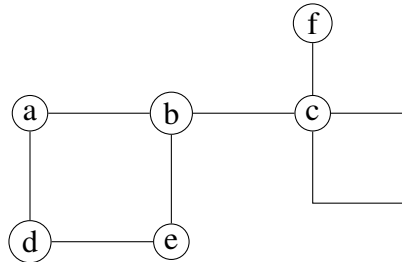
3. Are these all the same graph? How can we tell? (What does “same” mean?)



4. Below is a neighborhood: each vertex is an intersection, each edge is a street. How busy is each intersection?

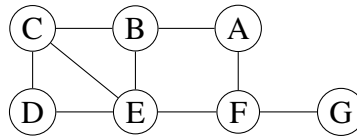
Degree

Loop degree



degree of a : _____ b : _____ c : _____ d : _____ e : _____ f : _____

5. Below is a road network: each vertex is a town, each edge is a road.



- (a) Can I get from here to there?

path:

Find a path from D to G : _____

- (b) Can I leave home and get back?

circuit:

Find a circuit starting at E : _____

- (c) Is everything reachable at all?

connected:

Which graphs are connected?

